📗 RS003 Week 3 — English Worksheet

Topic: Build the Turret Base · **Course:** Pygame Turret · **Time:** about 45 minutes

This week you draw the first piece of your turret: a **base** that sits in the middle-bottom of the screen. You learn to draw shapes, store important numbers as **constants**, and find the centre of the screen with **integer division** (`//`).

> 🧠 Words to know: **constant**, **rectangle**, **circle**, **integer division (`//`)**, **centre**

1 · Predict 🔮

Read each piece of code. Before you run it, write what you think will happen.

```
WIDTH = 800
CENTER_X = WIDTH // 2
```

What number is `CENTER_X`? Why use `//` instead of `/` here?

```
pygame.draw.rect(screen, GREEN, (CENTER_X - 30, HEIGHT - 80, 60, 40))
```

The four numbers are (left, top, width, height). Where on the screen does this rectangle appear, and how big is it?

```
pygame.draw.circle(screen, SILVER, (CENTER_X, HEIGHT - 90), 12)
```

What shape, where, and how big? What does the last number 12 control?

2 · Run It 🏃

🎯 Type the example, run it, and find the base

```
import pygame
pygame.init()

WIDTH = 800
HEIGHT = 600
CENTER_X = WIDTH // 2

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Turret Base")

SKY = (135, 206, 235)
DARK_GREEN = (0, 100, 0)

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SKY)
    pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 30, HEIGHT - 80, 60, 40))
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Where does the green rectangle sit? Is it really in the centre across? How can you tell?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The base should sit in the **centre** across the screen. Right now it is glued to the left edge.

```
pygame.draw.rect(screen, DARK_GREEN, (0, HEIGHT - 80, 60, 40))
```

Hint: use `CENTER_X` to place it in the middle.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — `CENTER_X` is supposed to be the middle of the screen, but it comes out as a number with a decimal point and the drawing looks off.

```
CENTER_X = WIDTH / 2
```

Hint: screen positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The base is supposed to sit **on the ground** near the bottom. Right now it is drawn near the top of the screen.

```
pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 30, 80, 60, 40))
```

Hint: in Pygame, Y = 0 is the **top**. The bottom is near `HEIGHT`.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It 🔧

🎯 Add a circle on top of the base

A turret needs a spinning joint. Add a small circle on top of the base where the gun will later sit.

```
pygame.draw.circle(screen, (180, 180, 180), (CENTER_X, HEIGHT - 80), 12)
```

Add this line below the rectangle, run it, and write what you see:

🎯 Resize the base using constants

Make the base wider, then narrower, by changing only the width number. Then try making the base its own constant so the size is easy to change in one place.

Write the constant you made and the size it sets (for example `BASE_WIDTH = 80`):

Hint: a constant is just a name in CAPITALS that holds a number you do not want to keep retyping.

5 · Make It 📸

🎯 **Draw your own turret base in the centre**

Build a window that:

1. uses constants for `WIDTH` and `HEIGHT`,
2. finds the centre with `//`,
3. draws a base rectangle in the middle-bottom,
4. adds a circle on top for the joint.

Send a **photo or video** of your base, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

> First, I set up my constants for ...
>
> I found the centre by ...
>
> I placed the base at the bottom by using ...
>
> I added a circle on top so that ...
>
> One tricky moment was when ...
>
> If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video of your turret base. Talk through it like you are teaching someone. Try to use these words: **constant**, **rectangle**, **circle**, **integer division**, **centre**.

1. Show your base sitting in the middle-bottom.
2. Read your `CENTER_X` line out loud and say what `//` does.
3. Point at the rectangle's four numbers and name each one.
4. Say why Y = 0 is the top, not the bottom.

Write what you will say in your video. Plan it here before you record.

Submit ✅

Send this worksheet + your photo or video to teacher on KakaoTalk.